

PANTEHA BEHBOODI

NEW YORK METROPOLITAN AREA

(201) 877-8345 Pbehbood@tyckolab.com
panteabehboodi1994@gmail.com
[Website](#) [LinkedIn](#) [Google Scholar](#)

I am a PhD biomedical scientist who translates novel therapeutics science into investment judgment.

PROFESSIONAL EXPERIENCE

Nucleate New York, Growth and Initiatives Team Lead

Aug 2026- Now

- Build out new verticals, educational programming, and events that extend Nucleate's reach beyond the core Activator program
- Design and run educational programs and event series (e.g., AI4Discovery, biohackathons, the Women in Life Sciences series)
- Roll out region-level pathway programming such as VC 101 and Consulting 101
- Develop challenges and collaborations with industry partners, including pharma
- Brainstorm and launch new initiatives across emerging verticals

H7 BioCapital, Business development fellow, Remote (SIE is scheduled for 10/2026)

March 2026- Now

- Conduct scientific due diligence on healthcare opportunities, evaluating intellectual property, competitive landscape, and market potential to support biotech and medtech investment decisions
- Prepare detailed investment recommendations for approval by the Investment Committee at H7 BioCapital

Center for Discovery and Innovation (CDI), HMMH, Post-doc fellow, Genetics and Epigenetics

May 2024- Now

- Conduct deep primary research on multiple myeloma biology, integrating genomics, epigenomics, and transcriptomics across ~150 patient samples to identify prognostic biomarkers and therapeutic liabilities
- Synthesize findings into clear written narratives for manuscripts, grant proposals, and conference presentations, translating complex science into decision-oriented insights
- Track daily literature, pipeline developments, and competitive landscape in hematologic malignancies
- Collaborate closely with clinicians, statisticians, and external academic partners, mirroring cross-functional coverage workflows

Regeneron Pharmaceuticals, Ph.D. R&D Co-Op, Human Systems

Jan 2023- Aug 2023

- Supported drug discovery and preclinical evaluation using a human Liver-Chip platform to assess therapeutic efficacy, toxicity, and biomarker responses in MAFLD and MASH
- Evaluated therapeutic candidates (Lipid nanoparticle-siRNA) by assessing target gene knockdown, pathway modulation, and disease-relevant readouts
- Generated and analyzed experimental datasets used to inform go/no-go decisions
- Communicated results to internal stakeholders through concise slide decks and data summaries
- Gained exposure to industry timelines, development risk, and translational decision-making

Stevens Institute of Technology, Ph.D. Researcher, Organ-on-Chip lab

Sep 2019- May 2025

- Led multiple research programs investigating mechanisms of drug resistance directly relevant to clinical outcomes and therapeutic durability in hematologic malignancies
- Designed physiologically relevant disease models to evaluate therapeutic response under clinically realistic conditions in multiple myeloma
- Integrated transcriptomic, proteomic, and functional data to support target prioritization
- Regularly presented findings to technical and non-technical audiences, strengthening investment-style communication skills

EDUCATION

Stevens Institute of Technology, Hoboken, NJ, Ph.D., Biomedical Engineering

May 2025

Thesis: "Optimizing multiple myeloma treatment approaches by elucidating tumor-stroma interactions in the bone marrow microenvironment." Identified key signaling pathways significantly enhancing drug resistance in MM within the BME and optimized therapeutic approaches.

Sharif University of Technology, Tehran, Iran, M.Sc., Materials Science and Engineering

May 2019

Thesis: "Design, Fabrication, and Improvement of Mechanical Properties and Biocompatibility of Hydroxyapatite-Chitosan Scaffold for Bone Tissue Engineering Applications."

Sharif University of Technology, Tehran, Iran, B.Sc., Materials Science and Engineering

May 2017

Thesis: "Development and Mechanical Properties of Bimodal Microstructures in Low Carbon Steel."

REFERENCES

Elias Pavlopoulos | Regeneron | Team Director & Co-op Mentor | ep@forthmanagement.com

Benjamin Tycko | Hackensack Meridian Health | Postdoctoral Advisor | BenjaminTycko@gmail.com

Woo Lee | Stevens Institute of Technology | PhD Advisor | wlee@stevens.edu

Marcin Iwanicki | Stevens Institute of Technology | Professor & Research Collaborator | Miwanick@stevens.edu

CERTIFICATIONS

[MIT Sloan School of Management](#), *Artificial Intelligence in Pharma and Biotech – Executive Program*

March 2026

Six-module executive program on applying AI and machine learning across the healthcare and drug development value chain: AI/ML foundations, AI-driven disease diagnosis and patient monitoring, NLP and data analytics in healthcare, interpretability in machine learning, patient risk stratification, and AI-enabled hospital management and optimization.

HONORS AND AWARDS

- 2025: New Jersey Health Foundation Research Grant Awardee (Grant # PC 238-25) – Awarded \$50K to support my PhD project titled “Understanding the Interactions Between Multiple Myeloma Cells and Osteoblasts to Optimize Treatment Approaches”
- 2025: [Johnson & Johnson Engineering Showcase Awardee](#) for 3-Minute Project Pitch
- 2025: UC Irvine Postdoctoral Fellowship Awardee (Relinquished)
- 2025: Weill Cornell Medicine Postdoctoral Fellowship Awardee (Relinquished)
- 2025: NIH Postdoctoral Fellowship Awardee (Relinquished)
- 2019: Provost Fellowship Awardee, the most prestigious PhD scholarship of Stevens Institute of Technology

PUBLICATIONS

Under Review

- Samimi S*, **Behboodi P***, Bariana M, Varkey A, Cassella E, Anuncio SA, Arun A, Siegel DS, Salas M, Castano A, Tycko B, Lee WY, Zakrzewski JL. Activation of the MIF–CD74–AKT Pathway by Endothelial Cells Promotes Myeloma Growth and Limits NK Cell–Mediated Killing. **HemaSphere**, submitted 2026. (*Co-first authors) · [Request draft](#)

Working Papers

- **Behboodi P***, Samimi S*, Castano A, Colorado I, Rigo M, Salas M, Tariq M, Feinman R, Zakrzewski JL, Lee W, Tycko B. Osteoblast–Myeloma Crosstalk Drives Trophic Interferon Pathway Signaling and Chemokines in Multiple Myeloma. *Co-first authors · Manuscript ready for submission · [Request draft](#)
- **Behboodi P**, Heduvor P, Pavlovič T, Salas M, Sarkar D, Tycko B, Iwanicki M. Chronic Low-Dose PFOS Exposure Alters the Transcriptome and DNA Methylome of Human Fallopian Tube Epithelial Cells. Manuscript ready for submission · [Request draft](#)
- **Behboodi P**, Alberca Doto C, Madrid A, Castano A, Salas M, Tariq M, Cummins A, Zhou A, Sarkar R, Kobbe R, Di Palma-Grisi J, Hillengass J, Slade M, Parekh S, Lagana A, Biran N, Chow K, Siegel D, Feinman R, Sucheston-Campbell L, Tycko B. Mechanisms and Biological Roles of Allele-Specific DNA Methylation in Multiple Myeloma. *Invited talk · ASH Annual Meeting, New Orleans, LA · December 2026 · [Request draft](#)*

Peer Reviewed

- Alshehri S, Pavlovič T, Farsinejad S, **Behboodi P**, et al. [Extracellular Matrix Modulates Outgrowth Dynamics in Ovarian Cancer](#). **Advanced Biology**, 2022.
- Arab M, **Behboodi P**, et al. [Enhanced Mechanical Properties and Biocompatibility of Hydroxyapatite Scaffolds by Magnesium and Titanium Oxides for Bone Tissue Applications](#). **Heliyon**, 2024.
- Iwanicki M, Pavlovic T, **Behboodi P**. [Engineering Approaches in Ovarian Cancer Cell Culture](#). Engineering and Physical Approaches to Cancer, Current Cancer Research, **Springer, Cham**, 2023.
- Aziz M, **Behboodi P**, Lee W. [Assessing Natural Killer Cell Cytotoxicity Against Multiple Myeloma Cells with IL-6 and sIL-6R](#). **Journal of Student Research**, 2024.
- Bokaii Hosseini Z, Rajabi F, Morovatshoar R, Ashrafpour M, **Behboodi P**, et al. [Downregulation of LPAR1 Promotes Invasive Behavior in Papillary Thyroid Carcinoma Cells](#). **Cancer Informatics**, 2024.

Conference Presentations and Workshops

- [MACROD2 and CDKN2A in Multiple Myeloma: Insights to Germline Susceptibility and Cytogenetic Risk from Long-Read Nanopore Sequencing](#). *ASCO Annual Meeting, Chicago · May 2026*
- [Effects of PFAS on DNA Methylation at Human Cancer Susceptibility Loci](#). *ASHG Annual Meeting, Boston · October 2025*
- [Optimizing Multiple Myeloma Treatment Approaches by Elucidating Tumor–Stroma Interactions in the Bone Marrow Microenvironment](#). *PhD thesis project · AACR Annual Meeting, Chicago · April 2025*
- Optimizing Multiple Myeloma Treatment. *Poster presentation · Johnson & Johnson Engineering Showcase, New Brunswick, NJ · April 2025*
- Allele-Specific Methylation Mapping and CRISPR Mutagenesis Identify Functional Chromatin-Regulating SNPs (rSNPs) in Disease Risk Loci. *Genetics and Epigenomics Research Conference, Center for Discovery and Innovation – Hackensack Meridian Health · 2024*
- Development of Human Systems for Genetic Disease Modeling, Drug Development and New Biology. *Human Systems, VelociGene · Regeneron Pharmaceuticals Inc., Tarrytown, NY · 2023*
- [Extracellular Matrix Modulates Outgrowth Dynamics in Ovarian Cancer](#). *48th Annual Northeast Bioengineering Conference, Columbia University, NY · 2022*
- Science Alliance Leadership Training (SALT) Workshop. *New York Academy of Sciences, NY · 2022*. Selected to participate in a comprehensive workshop focused on enhancing leadership skills.

MEMBERSHIPS

- American Society of Hematology (ASH)
- American Association for Cancer Research (AACR)
- American Society of Human Genetics (ASHG)
- American Society of Clinical Oncology (ASCO)